Curriculum for the Master's Degree Programme

Computational Social Systems

Curriculum 2021 in the version of 2025

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The Master's Degree Programme Computational Social Systems is a jointly offered degree programme (§ 51 (2) 27 Universities Act 2002) of the University of Graz (Uni Graz) and Graz University of Technology (TU Graz) within the framework of "Route 63". The legal bases of this degree programme are the Universities Act 2002 (UG) and the Legal Regulations for Academic Affairs in the statutes of TU Graz and Uni Graz, as amended.

(Please note: The English version of this document is a courtesy translation. Only the German version is legally binding.)

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I. General

§ 1 Subject matter of the degree programme and qualification profile

The Master's Degree Programme Computational Social Systems (abbrev. CSS) is an interdisciplinary degree programme. Graduates of this programme are awarded the university degree of "Master of Science," abbreviated: "MSc." The Master's Degree Programme Computational Social Systems is provided as a foreign-language degree programme in English.

Subject matter of the degree programme

The English-language Master's Degree Programme Computational Social Systems provides students with an interdisciplinary education at the interface of computer science, economics, sociology, psychology, and law. These subjects combined relate to socio-technical systems. On the path to a digital society, such systems are omnipresent and permeate all areas of political, economic, and academic life.

The functionality of computational social systems generally arises from the interaction of algorithms and the social behaviour of users: algorithms determine results based on the data generated by users' interactions with each other or with the system itself. Social, economic, legal, and technical issues are therefore generally very difficult to consider in isolation and an interdisciplinary perspective is required to evaluate these socio-technical systems.

Qualification profile and competences

The joint Master's Degree Programme Computational Social Systems aims to provide students with an interdisciplinary education at precisely these interfaces between relevant academic and scientific disciplines. This requires, on the one hand, academic training in data structures, algorithms, statistics, machine learning, data mining, and data science, and, on the other hand, an in-depth understanding of user behaviour, social and economic systems, their challenges, and their legal and ethical consequences.

(1) Graduates of the Master's Degree Programme Computational Social Systems have the following knowledge, skills and competencies:

Knowledge and understanding

Graduates

- have mastered relevant computer science methods, particularly in the areas of data science and machine learning.
- understand the fundamentals and key methods of business administration, psychology, law, or sociology depending on the chosen specialisation, and have introductory knowledge in all of these areas.

Application of knowledge

Graduates are able to

- analyse and design complex socio-technical systems systematically and in an interdisciplinary manner.
- design and conduct meaningful and methodically sound studies using modern computer science methods, as well as interpret the results obtained from these studies, incorporating theories from social science, economics, and cognitive science.
- apply their knowledge and skills to solve problems, including in new and unfamiliar situations.
- define and interpret the characteristics, limitations, terminologies, and schools of thought of at least two disciplines.

Evaluation and assessment

Graduates are able to

- analyse and evaluate the ethical, economic, technical, social, psychological, or legal aspects of socio-technical systems.
- formulate scientifically well-reasoned opinions and assess their resilience.

Communicative, organisational and social skills

Graduates

- are able to communicate in a subject-specific and goal-oriented manner with computer scientists and – depending on their chosen specialisation – with business economists, psychologists, sociologists, or legal scholars, and act as an interface between these professional groups.
- have mastered presentation techniques.
- are able to write academic and scientific texts.
- can independently acquire advanced knowledge.
- possess long-term relevant knowledge due to their broad education focused on fundamentals.

Need and relevance of the degree programme for science and the labour market

Graduates have the knowledge and skills to make significant, design-oriented contributions to the future development of society, for example, by providing new scientific insights into socio-technical systems, developing future technologies for them, or critically reflecting on their social impact.

Graduates fill a labour market need for interdisciplinary professionals who understand the specific languages and methods of the core disciplines covered in this degree programme. They are able to analyse social processes and systems using information technology tools and critically examine information technology projects using business, psychological, sociological, or legal approaches. They thus have the skills to steward the rapidly advancing digitalisation of society and to reflect on any developments critically and in an interdisciplinary manner.

Furthermore, graduates can be employed in fundamental and applied research on computational social systems. The master's degree programme also provides students with the skills required for independent scientific work within the framework of a doctoral programme.

§ 2 Admission requirements

(1) The following bachelor's and diploma degree programmes are eligible for admission to the joint Master's Degree Programme Computational Social Systems of Graz University of Technology and the University of Graz without further requirements: Computer Science, Business Administration, Sociology, Psychology, and Law. Furthermore, graduates of the Teacher Education Programme for the school subjects "Computer Science and Digital Basic Education" who completed the computer science programme or the school subject "Psychology/Philosophy" who completed the psychology programme, as well as graduates of the Bachelor's Degree Programme Law, Business and Society in Times of Digital Transformation who completed the law programme meet the admission requirements for the Master's Degree Programme Computational Social Systems.

(2) Any degree programmes that are not mentioned in (1) are considered eligible for admission if at least 60 ECTS credit points in each of the subject areas listed under (a.) to (e.) and at least one examination per subject area have been positively completed.

Degree programmes eligible for admission	**Subject areas**
a) Computer Science	• Fundamental mathematics • Software development • Information Processing • Theory and application of computer science
b) Business Administration	• Accounting • Finance • Marketing • Production and logistics • Management • Business mathematics and statistics • Economics – microeconomics, macroeconomics, international economics, public finance
c) Sociology	• History of sociology and sociological theory • Empirical social research • Micro-, meso-, and macrosociology and special sociologies • Statistics
d) Psychology	• Fundamentals (general psychology, biological psychology, neuropsychology, developmental psychology, differential psychology, social psychology) • Applied psychology (clinical psychology) • Health psychology, educational psychology, workplace and organisational psychology • Descriptive statistics, inferential statistics, psychological research methods • Psychological diagnostics
e) Law	• Civil law • Public law • Civil procedure law • European law • Criminal law • Corporate law

(3) Any degree programmes that are not mentioned in (1) and that do not meet the requirements of (2) are not considered equivalent to a subject-related degree programme. If at least 30 ECTS credit points have been completed in the subject areas mentioned in (2), full equivalence may be established by requiring supplementary examinations. Additional completion of supplementary examinations may be required to the extent of a maximum of 30 ECTS credit points.

(4) Any degree programmes that are not mentioned in (1) and do not meet the requirements of (2) and (3) are not close enough in subject matter to establish full equivalency, as they cannot establish equivalency even with supplementary examinations amounting to a maximum of 30 ECTS credit points. In such cases, admission to the Master's Degree Programme Computational Social Systems is not possible.

(5) Proof of sufficient English language skills is a prerequisite for admission to the degree programme. The type of proof required is to be specified in a regulation issued by the Rectorate.

§ 3 Structure of the degree programme

The Master's Degree Programme Computational Social Systems with a total workload of 120 ECTS credit points comprises four semesters and is structured into modules as follows:

	ECTS
1. Introduction to Computational Social Systems	Min. 42
2. Specialisations (one must be chosen) 2.A: Specialisation in Business Analytics 2.B: Specialisation in Societies, Technologies and Social Research 2.C: Specialisation in Human Factors 2.D: Specialisation in Law and Computer Science	Min. 42
Master's thesis colloquium	2
Master's thesis	27
Master's examination	1
Free-choice subjects (free electives)	Min. 6
Total	120

The degree programme phase "1. Introduction to Computational Social Systems" is designed to introduce students with different previously completed degree programmes to the field of computational social systems and to impart the skills they need for the interdisciplinary specialisations. Students can then focus on specific subject areas by choosing a specialisation. They further have to complete free-choice subjects (free electives), a master's thesis colloquium seminar, a master's thesis, and a master's examination to complete the degree programme.

Examinations taken as part of a bachelor's or diploma degree programme that served as a prerequisite for admission to a subsequent master's degree programme can only be recognised for this master's degree programme to the extent that the scope of the bachelor's or diploma degree programme exceeds 180 ECTS credit points.

§ 4 Group sizes

The following maximum numbers of participants (group sizes) have been established:

Lecture (VO) Lecture part of lecture with integrated exercises (VU)	no restriction
Exercise (UE) Exercise part of lecture with integrated exercises (VU)	30
Introductory seminar (PS)	25
Seminar (SE)	25
Classes (KS)	30
Design exercise (KU)	30

Deviating from this, the following maximum number of participants (group sizes) apply to the following modules/courses:

Some courses are taken from the curricula of other degree programmes (identifier other than “CSS”); the number of participants for such courses is defined in the curriculum of the respective degree programme.

§ 5 Guidelines for the allocation of places in courses

(1) If the number of students exceeds the number of available places, parallel courses are to be provided. If necessary, these parallel courses may also be provided during the holidays and semester breaks.

(2) The following applies to courses taken from other degree programmes: If it is not possible to offer a sufficient number of parallel courses (groups), the students are to be admitted to the course according to the priority order specified in the relevant curriculum. The currently valid version of the relevant degree programme curricula is applicable.

(3) The following applies to all other courses: If it is not possible to offer a sufficient number of parallel courses (groups), the students are to be admitted to the course according to the following priority ranking:

a. Position of the course in the curriculum (acc. to § 6): Priority is given to students for whom the course is compulsory according to their curriculum over those who are taking the course as part of an elective module or free-choice subject.
b. Total of completed/recognised ECTS credit points for the degree programme: All study achievements completed in the degree programme for which the student wants to take the course are taken into account for the ranking. Students with the highest total of ECTS credit points already completed in their current degree programme are ranked preferentially.
c. Number of semesters spent studying the degree programme so far: Students are ranked according to the number of semesters they have already studied in the degree programme. Priority is given to those who have studied for longer.

d. Decision by lot: If it is not possible to rank students according to the above criteria, admission to the course is decided by lot.

(4) Up to 10% of the existing places on the course are reserved for students completing part of their studies at TU Graz or the University of Graz as part of a mobility programme.

II. Degree Programme Content and Structure

§ 6 Modules, courses and semester assignment

The individual courses of this master's degree programme and their designation as compulsory and elective modules are set out below. The knowledge, methods or skills to be taught in each course are described in detail in Appendix I. The assignment of courses to specific semesters is a recommendation for students and ensures that the sequence of courses is best able to build on prior knowledge and that the workload of an academic year does not exceed 60 ECTS credit points. The courses are assigned to specific subject areas within the degree programme, which are offered by the participating universities; in the following tables, these subject areas are indicated in the column "Area."

Unless otherwise stated, lectures with integrated exercises (VU) are offered with 2/3 of the semester course hours held as lectures and 1/3 as exercises.

In order to ensure that a bachelor's degree programme and subsequent subject-related master's degree programme combined comprise a total workload of 300 ECTS credit points, one and the same course cannot be recognised for both the bachelor's degree programme qualifying the student for admission to a master's degree programme, and the master's degree programme in question. If a student has already completed a compulsory course of this master's degree programme within the framework of a previous degree programme qualifying the student for admission to this master's degree programme (e.g., bachelor's degree programme), the student must complete an additional elective course of the same scope to replace the already completed compulsory course in consultation with the relevant officer responsible for study matters.

Master's Degree Programme Computational Social Systems											
Module	**Course**				**Semester with ECTS credit points**				**Uni**	**TU**	**Area**
		SSt	**Type**	**ECTS**	**I**	**II**	**III**	**IV**			
1. Introduction to Computational Social Systems											
Module 1.1: General Introduction to Computational Social Systems											
Compul.	Introduction to Computational Social Systems	2	VU	5	5				x	x	CSS
Compul.	Foundations of Computational Social Systems	2	VU	5	5					x	CSS
Compul.	Research Design	2	VO	5	5				x		CSS
Subtotal Module 1.1		**6**		**15**	**15**						
Module 1.2: Introduction to Business Analytics for Students with a Computer Science Background											
Compul.	Introduction to Business Administration for Computer Science	2	KS	4	4				x		CSS
Elective	Masterkurs Accounting*	2	VO	4	4				x		BA
Elective	Master Course Finance	2	VO	4	4				x		BA

Master's Degree Programme Computational Social Systems											
Module	Course				Semester with ECTS credit points				Uni	TU	Area
		SSt	Type	ECTS	I	II	III	IV			
Elective	Master Course Marketing	2	VO	4	4				x		BA
Elective	Master Course Production and Logistics	2	VO	4	4				x		BA
Elective	Masterkurs Management*	2	VO	4	4				x		BA
Subtotal Module 1.2		**6**		**12**	**12**						
Module 1.3: Introduction to Societies, Technologies and Social Research for Students with a Computer Science Background											
Compul.	Introduction to Sociology	2	VO	4	4				x		SOC
Compul.	Introduction to Empirical Social Research	2	VO	3	3				x		SOC
Elective	Selected course from "Social Analysis"**		VU/ KS	0-8	0-8				x		SOC
Elective	Selected course from "Empirical Social Research"**		VU/ KS	0-10	0-10				x		SOC
Elective	Selected course from "Core Areas of Sociology"**		VU/ KS	0-8	0-8				x		SOC
Subtotal Module 1.3											
Module 1.4: Introduction to Human Factors for Students with a Computer Science Background											
Compul.	Introduction to Psychology I	2	VO	3	3				x		CSS
Compul.	Introduction to Psychology II	2	VU	3	3				x		CSS
Compul.	Selected course from "Topics in Psychology of Human Factors I"		SE/ VO/ VU	3-4	3-4				x		PSY
Elective	Selected course from "Topics in Psychology of Human Factors II"		SE/ VO/ VU	6-8	6-8				x		PSY
Subtotal Module 1.4				**min. 12**	**min. 12**						
Module 1.5: Introduction to Law and Computer Science for Students with a Computer Science Background											
Compul.	Introduction to Law	1	VU	1.5	1.5				x		LAW
Compul.	Computer Ethics	2	VU	3	3				x		CSS
Compul.	Technical Standards and Law	2	VO	5	5				x		LAW
Compul.	Legal Databases	1	KS	2.5	2.5				x		LAW
Subtotal Module 1.5		**6**		**12**	**12**						
Module 1.6: introduction to Computer Science for Students with a Background in Business Administration, Law, Sociology, Psychology and comparable											
Compul.	Informatics 1	3	VU	4						x	CS
Compul.	Databases	2	VU	3						x	CS
Compul.	Computational Methods for Statistics	2	VU	2.5						x	CS
Elective	Artificial Intelligence 1	2	VU	3						x	CS
Elective	Design Your Own App	2	VU	3						x	CS
Subtotal Module 1.6		**9**		**12.5**	**12.5**						
Module 1.7: Joint Module Computational Social Systems											
Compul.	Computational Modelling of Social Systems	3	VU	4.5		4.5				x	CS
Compul.	Knowledge Discovery and Data Mining 1	2	VO	3		3				x	CS
Compul.	Knowledge Discovery and Data Mining 1	1	KU	1.5		1.5				x	CS
Compul.	Introduction to IT Law	2	VO	3		3			x		LAW
Compul.	Social Aspects of Digital Technologies: Gender, Diversity and Research Ethics	2	VU	3	3				x		SOC

Master's Degree Programme Computational Social Systems											
Module	Course	SSt	Type	ECTS	Semester with ECTS credit points I	II	III	IV	Uni	TU	Area
Subtotal Module 1.7		**10**		**15**	**3**	**12**					
Total 1. Introduction to Computational Social Systems				**min. 42**	**min. 27**	**12**	**3**				
2. Specialisations											
2.A Specialisation in Business Analytics											
Module 2.A.1: Business Analytics											
Compul.	Data-Driven Business Model Seminar	2	PS	4		4			x	x	CSS
Compul.	Data Science in Business 1	2	PS	4			4		x	x	CSS
Compul.	Data Science in Business 2	2	PS	4			4		x	x	CSS
Subtotal Module 2.A.1		**6**		**12**		**4**	**8**				
Module 2.A.2: Elective Module Business Analytics											
Elective	Business Intelligence and Analytics	2	PS	4		4			x		BA
Elective	Data Management for Business Analytics	2	PS	4			4		x		BA
Elective	Data-Driven Decision Support	2	PS	4		4			x		BA
Elective	Case Studies Business Analytics	2	PS	4		4			x		BA
Elective	Specific Topics in Business Administration course (SBWL) from the Master's Degree Programme Business Administration: The four introductory seminars of SBWL, totalling 16 ECTS credit points, must be completed.**		PS	16		16			x		BA
Subtotal Module 2.A.2				**min. 12**		**min. 12**					
Module 2.A.3: Computer Science for Business Analytics											
Elective	Visual Analytics	3	VU	5		5				x	CS
Elective	Information Visualisation	3	VU	5		5				x	CS
Elective	Knowledge Discovery and Data Mining 2	3	VU	5			5			x	CS
Elective	Recommender Systems	2	VU	3		3				x	CS
Elective	Network Science	3	VU	5			5			x	CS
Elective	Social Media Technologies	2	VU	3		3				x	CS
Elective	Introduction to International Entrepreneurship	3	VU	5			5			x	CS
Elective	Information Search and Retrieval	3	VU	5			5			x	CS
Elective	Configuration Systems	2	VU	3			3			x	CS
Elective	Software Development Process*	1	VO	1.5		1.5				x	CS
Elective	Advanced Information Retrieval	3	VU	5			5			x	CS
Elective	Data Analysis and Introduction to R	2	VO	3			3			x	CS
Elective	Data Analysis and Introduction to R	1	UE	2			2			x	CS
Elective	Introduction to Data Structures and Algorithms	2	VO	3		3				x	CS
Elective	Introduction to Data Structures and Algorithms	1	UE	1.5		1.5				x	CS
Elective	Machine Learning 1	3	VU	4.5			4.5			x	CS
Elective	Human-Centred Design	2	VU	3		3				x	CS
Elective	Secure Product Lifecycle	2	VO	3			3			x	CS
Elective	Secure Product Lifecycle	1	KU	2			2			x	CS
Elective	Economic Systems	3	VU	5		5				x	CSS

Master's Degree Programme Computational Social Systems											
Module	Course	SSt	Type	ECTS	Semester with ECTS credit points I	II	III	IV	Uni	TU	Area
Elective	Computational Finance	3	VU	5			5			x	CSS
Subtotal Module 2.A.3				**min. 11**		**min. 11**					
Total 2.A Specialisation in Business Analytics				**42**		**42**					
2.B Specialisation in Societies, Technologies and Social Research											
Module 2.B.1: Societies, Technologies and Social Research											
Compul.	Research Seminar	3	KS	6			6		x	x	CSS
Compul.	Culture, Social Change, Technology	2	KS	4		4			x		SOC
Subtotal Module 2.B.1		**5**		**10**		**4**	**6**				
Module 2.B.2: Digital Societies: Theories, Methods and Critical Reflections											
Elective	Selected course from "Advanced Empirical Social Research"**		KS/ VU	0-21		0-21	0-21		x		SOC
Elective	Selected course from "Special Topics in Sociology"**		SE/ KS/ VU	0-21		0-21	0-21		x		SOC
Elective	Selected course from "Science and Technology Studies"		VO/ VU/ UE/ SE	0-21		0-21	0-21			x	CS
Subtotal Module 2.B.2											
Module 2.B.3: Computer Science for Societies, Technologies and Social Research											
Elective	Machine Learning 1	3	VU	4.5			4.5			x	CS
Elective	Introduction to Data Structures and Algorithms	2	VO	3		3				x	CS
Elective	Introduction to Data Structures and Algorithms	1	UE	1.5		1.5				x	CS
Elective	Knowledge Discovery and Data Mining 2	3	VU	5			5			x	CS
Elective	Natural Language Processing	3	VU	5		5				x	CS
Elective	Recommender Systems	2	VU	3		3				x	CS
Elective	Data Analysis and Introduction to R	2	VO	3			3			x	CS
Elective	Data Analysis and Introduction to R	1	UE	2			2			x	CS
Elective	Seminar Data Science	3	SE	5			5			x	CS
Elective	Information Search and Retrieval	3	VU	5			5			x	CS
Elective	Network Science	3	VU	5			5			x	CS
Elective	Visual Analytics	3	VU	5		5				x	CS
Elective	Social Media Technologies	2	VU	3		3				x	CS
Elective	Advanced Information Retrieval	3	VU	5			5			x	CS
Elective	Secure Product Lifecycle	2	VO	3			3			x	CS
Elective	Secure Product Lifecycle	1	KU	2			2			x	CS
Elective	Economic Systems	3	VU	5		5				x	CSS
Elective	Computational Finance	3	VU	5			5			x	CSS
Elective	Human-Centred Design	2	VU	3		3				x	CS
Subtotal Module 2.B.3				**min. 11**		**min. 11**					
Total 2.B Specialisation in Societies, Technologies and Social Research				**42**		**42**					

Master's Degree Programme Computational Social Systems											
Module	Course	SSt	Type	ECTS	Semester with ECTS credit points I	II	III	IV	Uni	TU	Area
2.C Specialisation in Human Factors											
Module 2.C.1: Human Factors Basics											
Compul.	Topics in Cognitive Psychology	2	VU	4		4			x		PSY
Compul.	Human-Centred Design	2	VU	3		3				x	CS
Compul.	Advanced Seminar (Decision-Making)	2	SE	4			4		x		PSY
Subtotal Module 2.C.1		**6**		**11**		**7**	**4**				
Module 2.C.2: Psychology of Human Factors											
Compul.	Empirical Seminar	4	SE	5		5			x		CSS
Elective	Selected course from "Modelling of Complex Systems"		VO/ VU/ SE/ KS/ PS/ UE	0-8			0-8		x		PSY
Elective	Selected course from "Topics in Robotics"			0-7			0-7		x		PSY
Elective	Selected course from "Quantitative Research Methods"			0-8		0-8	0-8		x		PSY
Elective	Selected course from "Special Topics in Psychology of Human Factors"			0-12		0-12	0-12		x		PSY
Subtotal Module 2.C.2				**min. 11**		**min. 11**					
Module 2.C.3: Computer Science for Human Factors											
Comp. El	Seminar Data Science	3	SE	5			5			x	CS
Comp. El	Seminar Interactive and Visual Information Systems	3	SE	5			5			x	CS
Elective	Human Computer Interaction	3	VU	4.5		4.5				x	CS
Elective	User Interfaces*	1.5	VU	2		2				x	CS
Elective	Human-AI Interaction 1	3	VU	5			5			x	CS
Elective	Visual Analytics	3	VU	5		5				x	CS
Elective	Information Visualisation	3	VU	5		3				x	CS
Elective	Game Design and Development I	3	VU	5			5			x	CS
Elective	Human-AI Interaction 2	2	VU	5		5				x	CS
Elective	Methods of Functional Brain Research	2	VO	3		3				x	CS
Elective	Cognitive Neuroscience	1	VO	3			3			x	CS
Elective	Data Analysis and Introduction to R	3	VO	3			3			x	CS
Elective	Data Analysis and Introduction to R	3	UE	2			2			x	CS
Elective	Information Search and Retrieval	3	VU	5			5			x	CS
Elective	Advanced Information Retrieval	3	VU	5			5			x	CS
Elective	Machine Learning 1	3	VU	4.5			4.5			x	CS
Elective	Network Science	3	VU	5			5			x	CS
Elective	Natural Language Processing	3	VU	5		5				x	CS
Elective	Knowledge Discovery and Data Mining 2	3	VU	5			5			x	CS
Elective	Secure Product Lifecycle	2	VO	3			3			x	CS
Elective	Secure Product Lifecycle	1	KU	2			2			x	CS
Elective	Economic Systems	3	VU	5		5				x	CSS
Elective	Computational Finance	3	VU	5			5			x	CSS
Subtotal Module 2.C.3				**min. 11**		**min. 11**					
Total 2.C Specialisation in Human Factors				**42**		**42**					

Master's Degree Programme Computational Social Systems											
Module	Course	SSt	Type	ECTS	Semester with ECTS credit points I	II	III	IV	Uni	TU	Area
2.D Specialisation in Law and Computer Science											
Module 2.D.1: Law by Design											
Compul.	Legal Lab 1	2	SE	6		6			x	x	CSS
Compul.	Legal Lab 2	4	SE	8			8		x	x	CSS
Subtotal Module 2.D.1		**6**		**14**		**6**	**8**				
Module 2.D.2 Fundamental Rights and the Data Economy											
Elective	Special Topics in Law and Computer Science 1	2	SE/ KS	5		5	5		x		LAW
Elective	Special Topics in Law and Computer Science 2	2	SE/ KS	5		5	5		x		LAW
Elective	Special Topics in Law and Computer Science 3	2	SE/ KS	5		5	5		x		LAW
Elective	Special Topics in Law and Computer Science 4	2	SE/ KS	5		5	5		x		LAW
Elective	Special Topics in Law and Computer Science 5*	2	SE/ KS	5		5	5		x		LAW
Elective	Special Topics in Law and Computer Science 6*	1	VO/ SE	1.5		1.5			x		LAW
Subtotal Module 2.D.2				**min. 10**		**min. 10**					
Module 2.D.3: Computer Science for Law and Computer Science											
Elective	Data Analysis and Introduction to R	2	VO	3			3			x	CS
Elective	Data Analysis and Introduction to R	1	UE	2			2			x	CS
Elective	Information Security	2.5	VO	4			4			x	CS
Elective	Information Security	2.5	KU	3			3			x	CS
Elective	Introduction to Data Structures and Algorithms	2	VO	3		3				x	CS
Elective	Introduction to Data Structures and Algorithms	1	UE	1.5		1.5				x	CS
Elective	Machine Learning 1	3	VU	4.5			4.5			x	CS
Elective	Visual Analytics	3	VU	5		5				x	CS
Elective	Knowledge Discovery and Data Mining 2	3	VU	5			5			x	CS
Elective	Recommender Systems	2	VU	3		3				x	CS
Elective	Artificial Intelligence 2	2	VU	3		3				x	CS
Elective	Information Search and Retrieval	3	VU	5			5			x	CS
Elective	Social Media Technologies	2	VU	3		3				x	CS
Elective	Advanced Information Retrieval	3	VU	5			5			x	CS
Elective	Natural Language Processing	3	VU	5		5				x	CS
Elective	Network Science	3	VU	5			5			x	CS
Elective	Economic Systems	3	VU	5		5				x	CSS
Elective	Computational Finance	3	VU	5			5			x	CSS
Elective	Human-Centred Design	2	VU	3		3				x	CS
Subtotal Module 2.D.3				**min. 10**		**min. 10**					
Total 2.D Specialisation in Law and Computer Science				**42**		**42**					
Master's thesis											
Master's thesis colloquium		**1**	**SE**	**2**			**2**		**x**	**x**	**CSS**
Master's thesis				**27**			**27**		**x**	**x**	**CSS**

Master's Degree Programme Computational Social Systems Module Course	SSt	Type	ECTS	Semester with ECTS credit points I	II	III	IV	Uni	TU	Area
Master's examination			1			1		x	x	CSS
Free-choice subjects (free electives)			min. 6					x	x	
Overall Total			12	30	30	30	30			

* This course is held in German.
** Includes courses in German and English.

§ 7 Free-choice subjects

(1) The courses to be completed as part of the free-choice subjects in the Master's Degree Programme Computational Social Systems are designed to provide individual strategic focus and further development of the students. They may be freely selected from the courses offered by any recognised national or international universities as well as recognised national post-secondary educational institutions. Appendix III contains recommendations for specific free-choice subjects.

(2) If no specific number of ECTS credit points is assigned to a free-choice subject, ECTS credit points are allocated by the officer responsible for study matters according to the estimated time spent on the course.

§ 8 Master's thesis

(1) The purpose of the master's thesis is to demonstrate a student's ability to work on scientific topics on their own, both with regard to content and methodology.

(2) The topic of the master's thesis must be an interdisciplinary topic taken from or meaningfully related to the chosen specialisation (2.A, 2.B, 2.C or 2.D) according to § 3.

(3) The master's thesis must be registered with the respective officer responsible for study matters via the Dean's Office before beginning work on it. The topic, the specialisation to which the topic is assigned, as well as the supervisor and associated institute need to be stated.

(4) All master's theses are co-supervised, with a primary supervisor from one of the participating universities and a co-supervisor from the other university.

(5) Students must present the research concept of their master's thesis in the master's thesis colloquium seminar before submitting the completed thesis.

§ 9 Registration requirements for courses/examinations

(1) The registration condition for the final master's examination before a committee is proof of positive assessment of all study achievements/examinations as described in § 6 to 7 as well as a positive grade for the master's thesis and the master's thesis colloquium.

(2) In addition, the listed registration requirements apply to the following courses/examinations/ modules: For Specialisations 2.A to 2.D, the modules listed in the table below apply as registration requirements in modules 2.A.1, 2.B.1, 2.C.1, and 2.D.1:

Module / Specialisation:	**Requirement**	
For Module 2.A.1 of the Specialisation Business Analytics	1.2	Introduction to Business Analytics for Students with a Computer Science Background or Prior degree in business administration according to § 2 (2) b
	and	
	1.6	Introduction to Computer Science for Students with Background in Business Administration, Law, Sociology, Psychology and comparable or Prior degree in computer science according to § 2 (2) a
For Module 2.B.1 of the Specialisation Societies, Technologies and Social Research	1.3	Introduction to Societies, Technologies and Social Research for Students with a Computer Science Background or Prior degree in sociology according to § 2 (2) c
	and	
	1.6	Introduction to Computer Science for Students with Background in Business Administration, Law, Sociology, Psychology and comparable or Prior degree in computer science according to § 2 (2) a
For Module 2.C.1 of the Specialisation Human Factors	1.4	Introduction to Human Factors for Students with a Computer Science Background or Prior degree in psychology according to § 2 (2) d
	and	
	1.6	Introduction to Computer Science for Students with Background in Business Administration, Law, Sociology, Psychology and comparable or Prior degree in computer science according to § 2 (2) a
For Module 2.D.1 of the Specialisation Law and Computer Science	1.5	Introduction to Law and Computer Science for Students with a Computer Science Background or Prior degree in law according to § 2 (2) e
	and	
	1.6	Introduction to Computer Science for Students with Background in Business Administration, Law, Sociology, Psychology and comparable or Prior degree in computer science according to § 2 (2) a

§ 10 Stays abroad and internships

(1) Recommended stays abroad

It is recommended for students to spend time abroad in the course of their studies. In this master's degree programme, the 3rd or 4th semesters are particularly suitable for this purpose.

It is also possible to obtain recognition of work done in shorter study periods abroad, for example participation in summer or winter schools, as part of the free-choice subjects, by application to the officer responsible for study matters.

(2) Internships

Students are strongly encouraged to complete a job-related internship within the framework of the free-choice subjects.

Each week of full employment corresponds to 1.5 ECTS credit points. Active participation in an academic event may also count as an internship. This internship must be approved by the officers responsible for study matters and considered a useful addition to the degree programme.

III. Examination Regulations and Conclusion of Studies

§ 11 Assessment of modules

The overall grade for a module is the average grade of all examinations completed as part of the module, weighted according to ECTS credit points. The grade is rounded up if the decimal place exceeds 0.5. Otherwise, the grade is rounded down. Examinations that are assessed only as "successfully completed/not completed" are not included in the calculation of the overall module assessment. Positive assessment of a module requires the positive assessment of all individual examinations to be completed within the module.

§ 12 Master's examination

(1) The master's examination is an oral examination before a committee and consists of

- the presentation of the master's thesis (max. 25 minutes), and
- the defence of the master's thesis (examination interview on the subject matter of the master's thesis and other subject-related areas).
- The defence of the master's thesis covers both the subject area of computer science and the subject area of the specialisation chosen for the master's thesis.

The total duration of the master's examination before a committee is usually 60 minutes and must not exceed 75 minutes.

(2) The members of the examination committee reflect the interdisciplinary nature of the degree programme. The examination committee for the master's examination includes of the supervisor and co-supervisor of the master's thesis and at least one other committee member who chairs the committee and is nominated by the relevant officer responsible for study matters, after hearing any recommendations from the candidate.

(3) The master's examination must be graded based on the performance achieved during the examination.

§ 13 Completion of studies

(1) The master's degree programme is completed once all academic achievements pursuant to § 3 have been assessed positively.

(2) Successful completion of the degree programme is documented by issuing a certificate. The master's degree certificate for the Master's Degree Programme Computational Social Systems is composed of:

a. a list of all completed modules set out in § 3 (along with their ECTS credit points) and their assessments,
b. the title and the assessment of the master's thesis,
c. the assessment of the master's thesis colloquium,
d. the assessment of the master's examination,
e. the total of the ECTS credit points of the free-choice subjects as defined in § 7, and
f. the overall assessment based on the assessment of the modules, the master's thesis, the master's examination and the master's thesis colloquium.

IV. Entry Into Force and Transitional Provisions

§ 14 Entry into force

This curriculum 2025 (abbrev. in UNIGRAZonline and TUGRAZonline "2025W") shall enter into force on 1 October 2025.

§ 15 Transitional provisions

Students of the Master's Degree Programme Computational Social Systems who are subject to the curriculum in the version from 2021 at the time when these changes to the curriculum come into force on 1 October 2025, will be subject to the 2025 version of the curriculum as of 1 October 2025.

Appendices to the Curriculum of the Master's Degree Programme Computational Social Systems

Appendix I: Module Descriptions

1. Introduction to Computational Social Systems

Module 1.1	**General Introduction to Computational Social Systems**
ECTS credit points:	15
Contents:	• Overview of the subject areas of Computational Social Systems, Business Analytics, Societies, Technologies and Social Research, Human Factors, and Law and Computer Science • Overview of methods: Qualitative, quantitative, experimental, and literature-based/hermeneutic research • Concepts of social systems and their relevance to information technology and digitalisation • Theoretical foundations of collective behaviour, social influence, emotional expression, and the structures of social networks • Methods for generating social data and information technology processes for such data • Statistical methods for testing hypotheses about social behaviour and understanding the behaviour of social systems
Learning outcomes:	After completing this module, students are able to • provide an overview of the subject areas covered in the degree programme. • describe various research methods. • fundamentally assess the potential and limitations of information technology methods for analysing human behaviour. • operationalise simple research questions about social systems that can be investigated using information technology methods. • integrate and process social data and model a small project on social data. • critically evaluate results and relate them to economic and social applications.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 1.2	**Introduction to Business Analytics for Students with a Computer Science Background**
ECTS credit points:	12
Contents:	Compulsory: • Fundamentals of business management and the tradition of business research • Basic knowledge of business problems, analysis methods, and solution approaches • Overview of the most important core areas of business administration Elective: • Accounting, finance, marketing, production and logistics, management

Learning outcomes:	Compulsory: After completing this module, students are able to • explain the fundamentals of business administration. • interpret and classify problems from a business administration perspective. • define and compare the core areas of business administration. Elective: After completing this module and depending on the chosen courses, students are able to • analyse and systematically solve simple problems from the areas of marketing, accounting, finance, or production and logistics. • analyse and systematically solve simple management problems.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 1.3	**Introduction to Societies, Technologies and Social Research for Students with a Background in Computer Science**
ECTS credit points:	Min. 12
Contents:	Compulsory: • Basic approaches and concepts of sociology • Relevance of social research for society • Procedures and methods of empirical social research • Sociological problems Elective: • Basic features of the social structure of Austrian society and its location in the international context; consideration of its historical development and the aspect of globalisation • Social problems of contemporary societies • Introduction to univariate, bivariate, and multivariate statistical data analysis and the use of statistical software; classification of empirical social research within sociology; connection of empirical social research with sociological theories • Core areas and interdependencies of micro-, meso-, and macrosociological phenomena as well as new developments in the respective fields • Microsociology: Focus on individuals in their immediate social context as well as small social groups; questions of the interaction between the individual and society; focus on face-to-face relationships • Mesosociology: Focus on the sociology of institutions and organisations, their internal structure and processes, as well as their interaction with individuals and society • Macrosociology: Focus on larger entities, such as societies or cultural groups; theories and empirical studies on phenomena of social change
Learning outcomes:	Compulsory: After completing this module, students are able to • identify basic terms and approaches in sociology and explain them in detail and in a differentiated manner. • theoretically describe the process of empirical research and different research designs, critically examine the most important methods of empirical social research and link them to specific questions.

	• identify the sociological aspects of a problem and distinguish them from other approaches, such as psychological or educational ones. • conduct an initial, rough analysis of social problems by applying the essential principles of sociological thought. Elective: After completing this module and depending on the chosen courses, students are able to • analyse Austrian society and its social structure from an international perspective. • apply their theoretical and methodological expertise to the diagnosis of social problems. • apply key methods of qualitative and quantitative social research and data analysis methods to social reality, interpret the results of empirical studies published in the specialist literature, and evaluate their methodological quality. • recapitulate basic concepts and fundamental questions of micro-, meso-, and macrosociology, as well as provide an overview of key positions in the three main areas and their representatives. • reflect on study results and the content of publications and apply them to developing one's own sociological research questions. • collaborate on content with others and subject it to critical reflection. • develop innovative and creative approaches based on sociological ideas and perspectives.
Prerequisites for participation:	Completion of the course "Elementary Data Analysis" is recommended before attending the course "Multivariate Data Analysis"
Frequency in which the module is provided:	Every academic year

Module 1.4	**Introduction to Human Factors for Students with a Computer Science Background**
ECTS credit points:	Min. 12
Contents:	Compulsory: • Introduction to the core areas of psychology, particularly human cognitive architecture • Introduction to human factors psychology Elective: • Selected chapters of human factors psychology
Learning outcomes:	Compulsory: After completing this module, students are able to • identify core topics in psychology. • interpret and classify problems from the perspective of human factors psychology. • define and compare the core areas of cognitive psychology. Elective: After completing this module and depending on the chosen courses, students are able to • critically evaluate fundamental research findings in psychology. • critically assess the most important methods of the relevant disciplines of psychology.
Prerequisites for participation:	none

Frequency in which the module is provided:	Every academic year

Module 1.5	**Introduction to Law and Computer Science for Students with a Computer Science Background**
ECTS credit points:	12
Contents:	• Fundamentals of normative sciences (ethics, law) as well as basic knowledge of technical standards, analytical methods, and solution approaches • Overview of the most important core normative areas
Learning outcomes:	After completing this module, students are able to • enumerate the fundamentals of normative sciences and technical standards. • interpret and classify problems from the perspective of ethics and law. • define and compare the core normative areas. • distinguish between moral norms, legal norms, and technical norms and classify them appropriately.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 1.6	**Introduction to Computer Science for Students with Background in Business Administration, Law, Sociology, Psychology and comparable**
ECTS credit points:	12.5
Contents:	• Introduction to the history of computer science, computer science terminology, software, and information systems • Programming methods and fundamentals of problem solving and programme modelling • Fundamentals of programming in Python with a focus on statistics and data processing • Knowledge representations and their use in machine learning • Fundamentals of information retrieval and recommendation systems • Database design and relational algebra, SQL queries and interfaces for application programming • Fundamentals of computer science methods for statistics, using software to perform simple significance tests
Learning outcomes:	After completing this module, students are able to • solve simple mathematical and practical problems using programming languages. • use computer science terminology and work in a secure computing environment. • explain data science and artificial intelligence and provide an overview of machine learning methods. • explain the functionality of database systems and apply this knowledge to data storage and access. • request and integrate data from various structured data sources. • create simple programmes that perform statistical analyses using computer science methods.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 1.7	Joint Module Computational Social Systems
ECTS credit points:	15
Contents:	• Fundamentals of representing human behaviour and social interaction in information technology models • Aggregated social dynamics and their calibration with empirical data • Basic concepts of social network dynamics • Concepts of dynamic systems applied to models of social interaction • Mathematical methods in data mining • Knowledge acquisition process • Text classification and grouping • Semantic analysis of text documents • Recommendation systems • Legal fundamentals for the use of information and communication technology • Selected data protection law, e-commerce law, copyright law, media law and computer criminal law case studies as well as Internet governance • Basics on the subject of fundamental rights and the Internet • Introductions to the following topics: Factors of inclusion and exclusion in the digital world, social inequality in access to and use of digital technologies, issues of gender gaps and diversity, as well as specific ethical issues in dealing with digital data, their analysis and interpretation
Learning outcomes:	After completing this module, students are able to • implement simple aspects of social and behavioural theory in algorithms. • implement individual aspects of human behaviour in computer science using Python. • mathematically analyse model dynamics and visualise this behaviour. • explain the mathematical foundations of data mining algorithms. • reflect on the necessary steps in the knowledge acquisition process. • identify data mining algorithms. • set up a simple knowledge acquisition project. • apply basic legal knowledge to the lawful handling and use of information and communication technology. • apply the acquired legal knowledge to comparable practical cases. • discuss central issues of inclusion and exclusion in the digital world in depth. • reflect on issues of gender and diversity and develop sensitivity for ethical issues in dealing with digital data, their analysis and interpretation.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

2. Specialisations

2.A Specialisation in Business Analytics

Module 2.A.1	Business Analytics
ECTS credit points:	12
Contents:	• Theories and technologies for data-based business models • Process models for developing data-based business models • Research methods from the field of business analytics • Research methods from the field of data science • Theories and technologies of interactive systems • Software development of interactive systems
Learning outcomes:	After completing this module, students are able to • select and apply suitable methods for the design and implementation of data-based business models. • develop a research design for a business analytics or data science problem. • select suitable methods for the design and implementation of interactive systems. • implement simple prototype solutions for business analytics problems. • assess the quality of business analytics solutions and, if necessary, conduct their own evaluation.
Prerequisites for participation:	Prerequisites for participation see § 9
Frequency in which the module is provided:	Every academic year

Module 2.A.2	Elective Module Business Analytics
ECTS credit points:	Min. 12
Contents:	Elective: • Concepts, technologies, and application of data-based decision support systems • Concepts, technologies, and application of business intelligence • Concepts, technologies, and application of data science from a business perspective • Concepts and technologies for managing large amounts of data • Content of a specialised business administration course
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • plan data-based decision support systems for business problems and select appropriate technologies. • develop a business intelligence concept for business problems. • create a data management concept for a business problem and select appropriate technologies. • implement simple prototype solutions for data management and decision-making support systems. • apply and implement theories and concepts to real-world problems within the subject area of a business intelligence course.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 2.A.3	Computer Science for Business Analytics
ECTS credit points:	Min. 11
Contents:	Elective: • Statistical analyses: fundamentals and implementation with real data, collection and processing of unstructured data on social interactions and behaviour • Basic concepts of machine learning and the use of machine learning in data analysis, text processing, and data collection • Theoretical and practical foundations of recommendation systems; methods for data visualisation and presenting insights based on data on social interactions and human behaviour • Modelling of social systems, social interaction, decision-making, and the structures of social systems • Technologies and interfaces for collecting and measuring data on social interactions and human behaviour • Data processing and collection of data from sociotechnical systems • Interfaces and systems for human interaction
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • analyse data sets and test hypotheses using statistical methods. • independently compile data sets from various social data sources. • process unstructured data using machine learning to quantify relevant variables of human behaviour. • understand recommendation systems and deal with information overload. • communicate knowledge generated from social and behavioural data. • identify the relevant technologies for measuring human behaviour, analysing data on social interactions and behaviours, and identifying technologies for interacting with people.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

2.B Specialisation in Societies, Technologies and Social Research

Module 2.B.1	Societies, Technologies and Social Research
ECTS credit points:	10
Contents:	The Research Seminar covers: • the exemplary application of previously acquired methodological and content-related knowledge. • the collaborative development of all phases of an empirical research project to be conducted independently by the students. • a comprehensive interdisciplinary examination of a specific topic. Culture, Social Change, Technology covers: • the terms, concepts, models, and theories of “culture,” “social change,” and “technology.” • the analysis of the interdependence of culture(s) and technology and the role of new technologies in modernisation processes. • the cultural interpretations of technology/progress/social change.

	• optimism about progress and criticism of civilization.
Learning outcomes:	After completing this module, students are able to • review and critically examine current literature and theory on the relationship between “culture,” “social change,” and “technology.” • understand the results of studies published in the specialist literature and evaluate their methodological quality, and based on this, contribute to the design of empirical studies. • conduct an empirical research project from the development of a research question to the presentation of the results in a small group. • apply various qualitative and/or quantitative research methods. • describe and present research results according to scientific standards. • work on complex tasks in small groups.
Prerequisites for participation:	Prerequisites for participation see § 9
Frequency in which the module is provided:	Every academic year

Module 2.B.2	**Digital Societies: Theories, Methods and Critical Reflections**
ECTS credit points:	Min. 12
Contents:	Elective: • Central concepts of the sociology of science and technology • Analysis and reflection of social and technological change • Science and technology as social institutions and social practices • Social constitution of scientific knowledge and technological artifacts • Increasing entanglement of science and technology with other social systems, such as the economy, politics, and the media • Philosophical and sociopolitical perspectives of modern information technologies and their applications • Technologies of self-measurement and body optimisation from a philosophical and sociological perspective • Historical development of feminist theory and gender theory as an analytical category in the social sciences • Key sociological areas such as family, physicality and sexuality, culture, education, work and institutions, and politics from a gender perspective • Selected fields of sociology and current sociological theories and models • Advanced quantitative methods of sociology: linear regression, variable transformations, and third-variable effects (confounding, interaction); logistic regression Cluster analysis, correspondence analysis • Advanced qualitative methods of sociology: basic concepts, procedures, and results of social network analysis; visual representation of networks; theoretical implications of the network concept for the study of social sciences
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • understand the social aspects of technology and science and address them as sociological questions. • gain an overview of the most important approaches in the sociology of science and technology and apply them to current problems. • develop an in-depth understanding of current literature in data science and establish connections to subject-related course content. • apply advanced quantitative and qualitative methods in specific projects.

	• independently review and critically examine current literature in the sociology of science and technology, and independently acquire further literature. • critically reflect on the pursuit of self-optimisation, analyse its normative implications, and interpret it against the background of philosophical and social science theories. • apply gender perspectives to the area of social phenomena. • critically discuss current sociological theories and models.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 2.B.3	**Computer Science for Societies, Technologies and Social Research**
ECTS credit points:	Min. 11
Contents:	Elective: • Fundamentals of statistical analysis and conducting analyses with “real” datasets • Data retrieval and processing of unstructured social and behavioural data • Basic concepts of machine learning and implementation of machine learning methods for data analysis, including text processing and data mining • Theoretical and practical principles of recommendation systems • Methods for data visualisation and presenting insights based on social and behavioural data • Data-generating technologies: • Modelling of social systems, including social interaction, decision-making, and social network structures • Measurement technologies and interfaces for capturing human behaviour and social interaction • Data processing and retrieval of existing data in computational social systems; interface and interactive system design for interaction with individuals
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • generate their own datasets from various social data sources. • process unstructured data using machine learning methods to quantify relevant variables related to human behaviour. • develop a basic understanding of the use of recommendation systems and technologies to cope with information overload. • understand the results of studies published in the specialist literature and evaluate them with regard to their methodological quality.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

2.C Specialisation in Human Factors

Module 2.C.1	Human Factors Basics
ECTS credit points:	10
Contents:	• Selected chapters of cognitive psychology • Methods and technologies for designing and developing interactive computer systems • Current topics in decision research, insights from behavioural and neuroscientific studies
Learning outcomes:	After completing this module, students are able to • identify core topics in cognitive psychology and provide an overview of the current state of theories, models, and methods. • develop a research design for a decision research project. • explain and discuss activity theory in the context of human-computer interfaces (HCI). • understand and apply contextual design methods and, within this context, design and develop innovative interactive systems.
Prerequisites for participation:	Prerequisites for participation see § 9
Frequency in which the module is provided:	Every academic year

Module 2.C.2	Psychology for Human Factors
ECTS credit points:	Min. 11
Contents:	Compulsory: • Empirical seminar: Extensive empirical work in the PSYLAB on a psychological question Elective: • Modelling complex systems: Modelling complex psychological system dynamics with computers • Selected robotics topics: e.g., historical background of artificial life, Turing machines, pattern-forming processes, multi-agent systems, evaluation of robotics experiments • Quantitative research methods: Mathematical methods for modelling trajectories in complex environments • Selected areas of human factors psychology
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • conduct an empirical study in human factors psychology. • handle the principles of artificial life and robotics. • assess the effects of behaviour-controlling feedback and understand the course of adaptation processes. • identify key aspects of complex systems in nature and their modelling, and explain optimisation methods based on natural phenomena. • reflect on test and item construction at an advanced level. • effectively apply the acquired knowledge of human factors psychology, for example, to design and evaluate a neurofeedback-based intervention. • critically reflect on the most important theories and empirical findings learned and assess their scientific and practical significance.
Prerequisites for participation:	none

Frequency in which the module is provided:	Every academic year

Module 2.C.3	**Computer Science for Human Factors**
ECTS credit points:	Min. 11
Contents:	Elective: • Introduction to human-computer interaction, with a focus on user interface design and usability engineering methods • Concepts of interfaces for human-computer interaction; user interface as an extension of human capabilities, with a focus on the theory of human perception and action, visual, auditory, olfactory, and haptic user interfaces • Introduction to the topics of human-aware interaction, concepts and principles of interfaces that utilise machine intelligence • Methods and techniques used in the design and implementation of wearable computing devices • Design of visual user interfaces for data analysis, techniques for the visual analysis of temporal, multidimensional, textual, and network data • Methods and skills required for game development, creation of 3D objects and environments, as well as the fundamentals of game design • Concepts and statistical methods using the R programming language • General introduction to brain signals (electrical, magnetic, metabolic) and measurement methods (invasive and non-invasive), experimental strategies, hardware, visualisation, real-time processing and signal processing (parameter extraction, classification, optimisation), applications, and functional brain research • Overview of selected theory and practice of information search and retrieval • Concepts and methods from the field of machine learning • Concepts and methods from the field of network science, the function and structure of complex networks, routing functions in complex networks, decentralised search, epidemics in complex networks, and models of information dissemination • Text analysis with a focus on written text, introduction to algorithms that enable the extraction of structured information from unstructured, textual resources • Overview of all elements of a secure product life cycle and in-depth study of selected topics
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • recapitulate the theory of human-computer interaction, design usable interfaces, and conduct heuristic evaluations and think-aloud tests of user interfaces. • design and develop the hardware, software drivers, and a prototype of a user interface. • design and implement visualisations and visual analysis systems. • design and implement wearable computing devices that incorporate some form of data processing as well as sensing and/or feedback. • explain methods of game design and development. • identify basic methods of functional brain research. • handle concepts and techniques in the field of cognitive neuroscience.

	• conduct analyses of real-world data problems, applying statistical techniques that enable the evaluation and inferential interpretation of individual effects. • explain and apply the following concepts: (1) the basic model of modern information retrieval, (2) the processing chain from text documents to search index, (3) performance characteristics, (4) the architecture of web search services, and (5) conduct literature reviews on selected topics in the field of information retrieval. • apply the most important methods of machine learning to practical problems. • explain and compare the relationship between the function and structure of complex networks and the common scientific methods in network science. • explain areas such as automatic text analysis, general understanding of natural language text, and the application areas of natural language processing. • reflect on the concept and importance of a secure product lifecycle.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

2.D Specialisation in Law and Computer Science

Module 2.D.1	**Law by Design**
ECTS credit points:	14
Contents:	• In-depth study and application of methodologies in computer science and law • Interdisciplinary application of legal norms in computer science in the form of legal prototyping in micro-projects
Learning outcomes:	After completing this module, students are able to • apply and implement theories and concepts of computer science and legal application to real-world problems in micro-projects. • implement the legal requirements of an IT system within the system in interdisciplinary micro-projects (law/computational science). • implement legal requirements for specific IT systems using simple prototype solutions. • assess the compliance of IT products and, if necessary, conduct their own evaluation.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Module 2.D.2	**Fundamental Rights and the Data Economy**
ECTS credit points:	min. 10
Contents:	• Theories of fundamental and human rights • Conventions and legal instruments (especially the ECHR and the Charter of Fundamental Rights) • Application principles of human and fundamental rights in specific IT situations • Case law on fundamental and human rights (ECJ and ECHR) • Artificial intelligence and discrimination

	• Data protection • Foundations of the Digital Single Market • Data markets and antitrust law • Intellectual property • Other relevant legal courses
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • understand and differentiate the concepts of fundamental and human rights. • identify the legal foundations of fundamental and human rights and the relevant decisions of the ECJ and ECHR in the context of computer science. • apply fundamental and human rights in specific scenarios. • identify and describe the data protection requirements for a system. • evaluate the significance of an IT product in the Digital Single Market. • identify the factors relevant to liability in an IT product. • apply the relevant standards of antitrust, liability, and licensing law, taking a justifiable balance of fundamental rights interests into account.
Prerequisites for participation:	Prerequisites for participation see § 9
Frequency in which the module is provided:	Every academic year

Module 2.D.3	**Computer Science for Law and Computer Science**
ECTS credit points:	min. 10
Contents:	Elective: • Advanced study in computer science with a focus on data analysis and its application in data-driven applications • Theories and fundamentals in the areas of algorithms, data analysis, machine learning, and key topics in information security • Application-oriented courses on various aspects of information systems
Learning outcomes:	After completing this module and depending on the chosen courses, students are able to • analyse basic algorithms, their assumptions, prerequisites, and behaviour. • review the fundamentals of data analysis, including statistical methods, machine learning, and visual methods. • identify suitable methods for analysing existing data and data sources and apply them in practical scenarios. • review search and recommendation technologies and assess their implications for existing use cases. • analyse complex network structures. • handle user-generated data and analyse social media. • identify and apply suitable methods for textual data sources.
Prerequisites for participation:	none
Frequency in which the module is provided:	Every academic year

Appendix II: Recommended Curriculum Timeline

The following curriculum timeline (example of a student with a previous degree in Computer Science who chose the Specialisation in Business Analytics) is not mandatory, but merely a recommendation and a guide for students.

Recommended Curriculum Timeline						
Module	**Course**	**SSt**	**Type**	**ECTS**	**Uni Graz[1]**	**TU Graz[1]**
1st semester						
1.1	Introduction to Computational Social Systems	2	VU	5	x	x
1.1	Foundations of Computational Social Systems	2	VU	5		x
1.1	Research Design	2	VO	5	x	
1.2	Introduction to Business Administration for Computer Scientists	2	KS	4	x	
1.1	e.g. Master Course Finance	2	VO	4	x	
1.1	e.g. Masterkurs Marketing	2	VO	4	x	
1.7	Social Aspects of Digital Technologies: Gender, Diversity and Research Ethics	2	VU	3	x	
Subtotal 1st semester		14		30		
2nd semester						
1.7	Computational Modelling of Social Systems	3	VU	4.5		x
1.7	Knowledge Discovery and Data Mining 1	2	VO	3		x
1.7	Knowledge Discovery and Data Mining 1	1	KU	1.5		x
1.7	Introduction to IT Law	2	VO	3	x	
2.A.1	Data-Driven Business Model Seminar	2	PS	4	x	x
2.A.2	e.g. Business Intelligence and Analytics	2	PS	4	x	
2.A.2	e.g. Data-Driven Decision Support	2	PS	4	x	
2.A.3	e.g. Visual Analytics	3	VU	5		x
Subtotal 2nd semester		17		30		
3rd semester						
2.A.1	Data Science in Business 1	2	PS	4	x	x
2.A.1	Data Science in Business 1	2	PS	4	x	x
2.A.2	e.g. Data Management for Business Analytics	2	PS	4	x	
2.A.3	e.g. Network Science	3	VU	5		x
2.A.3	e.g. Introduction to International Entrepreneurship	3	VU	5		x
2.A.3	e.g. Secure Product Lifecycle	2	VO	3		x
	Free-choice subjects (free electives)			6	x	x
Subtotal 3rd semester		14		31		
4th semester						
	Master's thesis			27		
	Master's thesis colloquium	1	SE	2		
	Master's examination			1		
Subtotal 4th semester		1		30		
Overall Total				**120**		

[1]: Assignment of the course to the participating universities. Both universities are named if the course is offered at both universities in combination, in parallel or alternately.

Appendix III: Recommended Free-Choice Subjects

As free-choice subjects, students can choose courses as desired in accordance with § 7 of this curriculum.

For students to broaden their knowledge, courses in the fields of foreign languages, social competence, technological impacts assessment and women's and gender studies are recommended. In particular, the following institutions and service departments are offered:

- Languages, Key Competencies and In-House Training of TU Graz,
- Science, Technology and Society Unit (STS Unit) of TU Graz,
- treffpunkt sprachen – Centre for Language, Plurilingualism and Didactics,
- the transfer initiative for management and entrepreneurship fundamentals, awareness, training and employability ("TIMEGATE"), and
- Centre for Social Competence of Uni Graz.

Appendix IV: Equivalence List

(1) Equivalency of courses when switching from the expiring Master's Degree Programme Computational Social Systems curriculum in the version of 2021 to the new curriculum version 2025

The courses of the new curriculum are listed on the left-hand side of the table. The corresponding equivalent courses in the expiring curriculum version of the Master's Degree Programme Computational Social Systems are listed on the right-hand side of the table. Any courses of the expiring curriculum that have no equivalent course according to this list may be taken as free-choice subjects. Courses that have the same name and type, number of ECTS credit points and the number of semester hours are considered to be equivalent and are thus not explicitly listed in the equivalence list.

Curriculum Computational Social Systems in the version of 2025					**Expiring curriculum Computational Social Systems in the version of 2021**				
	Course	Type	ECTS	SSt		Course	Type	ECTS	SSt
1.6	Artificial Intelligence 1	VU	3	2	1.6	Introduction to Data Science and Artificial Intelligence	VU	3	2
2.A.3 2.B.3 2.C.1 2.D.3	Human-Centred Design	VU	3	2	2.A.3 2.C.1	Designing Interactive Systems	VU	3	2
2.A.3 2.B.3 2.D.3	Introduction to Data Structures and Algorithms	VO	3	2	2.A.3 2.B.3 2.D.2	Data Structures and Algorithms 1	VO	3	2
2.A.3 2.B.3 2.D.3	Introduction to Data Structures and Algorithms	UE	1.5	1	2.A.3 2.B.3 2.D.2	Data Structures and Algorithms 1	UE	1.5	1
2.A.3 2.B.3 2.D.3	Machine Learning 1	VU	3	1	2.A.3	Machine Learning 1	VO	3	2
					2.B.3 2.C.3 2.D.2	Machine Learning 1	UE	1.5	1
2.A.3 2.B.3 2.C.3 2.D.3	Economic Systems	VU	5	3		no equivalence			
2.A.3 2.B.3 2.C.3	Computational Finance	VU	5	3		no equivalence			

Curriculum Computational Social Systems in the version of 2025					Expiring curriculum Computational Social Systems in the version of 2021				
	Course	Type	ECTS	SSt		Course	Type	ECTS	SSt
2.D.3									
2.B.3 2.C.3	Seminar Data Science	SE	5	3	2.C.3	Human Factors Lab	SE	5	4
2.C.3	Seminar Interactive and Visual Information Systems	SE	5	3					
2.C.3	Human-AI Interaction 1	VU	5	3	2.C.3	Intelligent User Interfaces	VU	5	3
2.C.3	Human-AI Interaction 2	VU	5	3	2.C.3	Wearable Computing	VU	5	3
2.D.1	Legal Lab 1	SE	6	2	2.D.3	Legal Lab 1	SE	6	2

(2) Regulations for continuing with the expiring Master's Degree Programme Computational Social Systems in the version of 2021

The courses of the expiring Master's Degree Programme Computational Social Systems are listed on the left-hand side of the table. On the right side of the table is a list of courses from the new curriculum that may be completed instead of the courses originally listed in the curriculum if the student wishes to remain in the expiring curriculum and the original courses are no longer offered. Courses that have the same name and type, number of ECTS credit points and the number of semester hours are considered to be equivalent and are thus not explicitly listed in the equivalence list.

Expiring curriculum Computational Social Systems in the version of 2021					Curriculum Computational Social Systems in the version of 2025				
	Course	Type	ECTS	SSt		Course	Type	ECTS	SSt
1.6	Introduction to Data Science and Artificial Intelligence	VU	3	2	1.6	Artificial Intelligence 1	VU	3	2
2.A.3 2.C.1	Designing Interactive Systems	VU	3	2	2.A.3 2.B.3 2.C.1 2.D.3	Human-Centred Design	VU	3	2
2.A.3 2.B.3 2.D.2	Data Structures and Algorithms 1	VO	3	2	2.A.3 2.B.3 2.D.3	Introduction to Data Structures and Algorithms	VO	3	2
2.A.3 2.B.3 2.D.2	Data Structures and Algorithms 1	UE	1.5	11	2.A.3 2.B.3 2.D.3	Introduction to Data Structures and Algorithms	UE	1.5	1

Expiring curriculum Computational Social Systems in the version of 2021					Curriculum Computational Social Systems in the version of 2025				
	Course	Type	ECTS	SSt		Course	Type	ECTS	SSt
2.A.3	Machine Learning 1	VO	3	22	2.A.3 2.B.3 2.C.3 2.D.3	Machine Learning 1	VU	3	1
2.B.3 2.C.3 2.D.2	Machine Learning 1	UE	1.5	1					
2.C.3	Human Factors Lab	SE	5	4	2.B.3 2.C.3	Seminar Data Science	SE	5	3
					2.C.3	Seminar interactive and Visual Information Systems	SE	5	3
2.C.3	Intelligent User Interfaces	VU	5	3	2.C.3	Human-AI Interaction 1	VU	5	3
2.C.3	Wearable Computing	VU	5	3	2.C.3	Human-AI Interaction 2	VU	5	3
2.C.3	Evaluation Methodology	VU	3	2		individual recognition			
2.D.3	Legal Lab 1	SE	6	2	2.D.1	Legal Lab 1	SE	6	2